

◆ Basic Git Interview Questions

1. What is Git?

Answer:

Git is a distributed version control system used to track changes in code, collaborate with teams, and manage project versions efficiently.

2. Difference between Git and GitHub?

Answer:

- **Git** → Version control tool (local system)
 - **GitHub** → Cloud platform to host Git repositories
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3. What is a repository?

Answer:

A repository (repo) is a folder where your project files and version history are stored.

4. What is the difference between `git pull` and `git fetch`?

Answer:

- `git fetch` → Downloads changes but doesn't merge
 - `git pull` → Downloads + merges changes into current branch
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5. What is a commit?

Answer:

A commit is a snapshot of changes in the project at a specific time.

◆ Intermediate Questions

6. What is the difference between `git merge` and `git rebase`?

Answer:

- **Merge** → Combines branches, keeps history
 - **Rebase** → Rewrites history, cleaner linear timeline
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7. What is a branch in Git?

Answer:

A branch is an independent line of development used to work on features without affecting main code.

8. What is a merge conflict?

Answer:

Occurs when Git cannot automatically merge changes due to conflicting edits in the same file.

9. What is `git stash`?

Answer:

Temporarily saves uncommitted changes so you can switch branches without committing.

10. What is `.gitignore`?

Answer:

A file that tells Git which files/folders to ignore (e.g., `node_modules`, `.env`).

◆ Advanced Questions (Important for Interviews)

11. What is HEAD in Git?

Answer:

HEAD is a pointer to the current commit or branch you are working on.

12. Difference between `git reset` and `git revert`?

Answer:

- `reset` → Deletes commits (history change)
 - `revert` → Creates new commit to undo changes (safe for shared repo)
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13. What is CI/CD in Git context?

Answer:

Using Git with pipelines to automatically test and deploy code (e.g., GitHub Actions).

14. How do you resolve merge conflicts?

Answer:

1. Open conflicting file
 2. Remove conflict markers (`<<<<`, `====`, `>>>>`)
 3. Keep correct code
 4. `git add + git commit`
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15. What is a pull request (PR)?

Answer:

A request to merge code from one branch into another (used for code review).

◆ Scenario-Based Questions (Very Important 🔥)

16. You pushed wrong code to main branch. What will you do?

Answer:

- Use `git revert` (safe way)
- Or `git reset + force push` (only if allowed)

17. How do you manage version control in your project?

Answer (your profile-based):

In my projects like JobVerse and LinkChat, I used Git with feature branches, pull requests, and proper commit messages. I followed a workflow where each feature was developed in a separate branch and merged after testing.

18. How do you collaborate with a team using Git?

Answer:

- Use branches (feature, dev, main)
 - Create PRs
 - Code reviews before merging
 - Resolve conflicts properly
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19. What is the difference between fork and clone?

Answer:

- **Fork** → Copy repo on GitHub account
 - **Clone** → Copy repo to local system
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20. What are good commit message practices?

Answer:

- Clear and short
 - Use present tense
 - Example:
 - 👉 Add user authentication API
 - 👉 Fix login bug
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 **Pro Tip (For Your Interview)**

Since your resume already includes **Git + real projects**, say this:

“I use Git daily for version control, follow branching strategies, and manage collaboration through pull requests and code reviews.”
